Akoizo Original Practice - Form 1 - Answer key

Division B | Dynamic Planet | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) One mm over one km^2 equals…

Unit: Water movement and water budgets.
Worked-case data: Rainfall is 20 mm over 2 km^2 with runoff fraction 0.30. Find direct runoff.

A. 1000 m^3

B. 1 m^3

C. 1 million m^3

D. 1000 liters

Correct: A

2. (2 points) Two second-order streams create…

Unit: Watersheds, stream order, and drainage patterns.
Worked-case data: A first-order stream joins a third-order stream, then another third-order joins. Find successive orders.

A. First order

B. Third order

C. Second order

D. Fourth order

Correct: B

3. (2 points) An oxbow commonly forms through…

Unit: Fluvial landforms and river history.
Worked-case data: Two events share peak discharge but one lasts longer. Why might their geomorphic effects differ?

A. Only coral growth

B. Only groundwater pumping

C. Meander cutoff

D. Only glacier freezing

Correct: C

4. (2 points) High porosity always means high permeability?

Unit: Groundwater, aquifers, karst, and Darcy flow.
Worked-case data: K=0.001 m/s, A=10 m^2, head drop=2 m over 100 m. Find Q.

A. Always

B. Only for clay

C. Only below water table

D. No

Correct: D

5. (2 points) Impervious connectivity can affect…

Unit: Land use, water pollution, and hydrological impacts.
Worked-case data: Concentration is 2 mg/L and discharge is 3 m^3/s. Find load rate.

A. Runoff timing

B. Only mineral hardness

C. Only gene frequency

D. No water process

Correct: A

6. (2 points) A budget requires…

Unit: Water movement and water budgets.
Worked-case data: Rainfall is 20 mm over 2 km^2 with runoff fraction 0.30. Find direct runoff.

A. Only a map color

B. Only a starting temperature

C. A boundary and interval

D. Only a river name

Correct: C

7. (2 points) Groundwater divides always match surface divides?

Unit: Watersheds, stream order, and drainage patterns.
Worked-case data: A first-order stream joins a third-order stream, then another third-order joins. Find successive orders.

A. Always

B. Only on maps

C. Only in winter

D. No

Correct: D

8. (2 points) Terrace elevation alone establishes exact age?

Unit: Fluvial landforms and river history.
Worked-case data: Two events share peak discharge but one lasts longer. Why might their geomorphic effects differ?

A. No

B. Always

C. Only for sandstone

D. Only below sea level

Correct: A

9. (2 points) Darcy flux equals…

Unit: Groundwater, aquifers, karst, and Darcy flow.
Worked-case data: K=0.001 m/s, A=10 m^2, head drop=2 m over 100 m. Find Q.

A. Porosity alone

B. Q/A

C. Q x A

D. K/A

Correct: B

10. (2 points) Load rate needs…

Unit: Land use, water pollution, and hydrological impacts.
Worked-case data: Concentration is 2 mg/L and discharge is 3 m^3/s. Find load rate.

A. Only map area

B. Only color

C. Concentration and discharge

D. Only pH

Correct: C

11. (2 points) Infiltrated water can later…

Unit: Water movement and water budgets.
Worked-case data: Rainfall is 20 mm over 2 km^2 with runoff fraction 0.30. Find direct runoff.

A. Support streamflow

B. Never move again

C. Only evaporate instantly

D. Create salt by definition

Correct: A

12. (2 points) Radial drainage can suggest…

Unit: Watersheds, stream order, and drainage patterns.
Worked-case data: A first-order stream joins a third-order stream, then another third-order joins. Find successive orders.

A. No topography

B. An elevated central feature

C. Only flat sediment

D. Only a straight fault

Correct: B

13. (2 points) Migration evidence is strongest from…

Unit: Fluvial landforms and river history.
Worked-case data: Two events share peak discharge but one lasts longer. Why might their geomorphic effects differ?

A. One water sample

B. Only river name

C. Comparable positions over time

D. One unlabeled image

Correct: C

14. (2 points) Flow generally follows…

Unit: Groundwater, aquifers, karst, and Darcy flow.
Worked-case data: K=0.001 m/s, A=10 m^2, head drop=2 m over 100 m. Find Q.

A. Only surface slope

B. Increasing head always

C. Only north

D. Decreasing hydraulic head

Correct: D

15. (2 points) Before/after change proves causality alone?

Unit: Land use, water pollution, and hydrological impacts.
Worked-case data: Concentration is 2 mg/L and discharge is 3 m^3/s. Find load rate.

A. No

B. Always

C. Only after one storm

D. Only in a reservoir

Correct: A

16. (4 points) What is missing from a simple P-Q budget?

Unit: Water movement and water budgets.
Worked-case data: Rainfall is 20 mm over 2 km^2 with runoff fraction 0.30. Find direct runoff.

answer: Evapotranspiration, storage change, and other boundary transfers may matter. (4 points).

17. (4 points) Why specify the outlet?

Unit: Watersheds, stream order, and drainage patterns.
Worked-case data: A first-order stream joins a third-order stream, then another third-order joins. Find successive orders.

answer: It defines the contributing basin and network considered. (4 points).

18. (4 points) What can form a point bar?

Unit: Fluvial landforms and river history.
Worked-case data: Two events share peak discharge but one lasts longer. Why might their geomorphic effects differ?

answer: Deposition on the inner part of a bend under suitable flow conditions. (4 points).